OpenADMET Blind Challenge
Predicting CYP Inhibition
Pico Methodology Report

11/3/2026

The intent of this exercise was to predict the direct inhibition pIC50 for CYP1A2, CYP2C9, CYP2D6 and CYP3A4 on a blinded set of 750 molecules, as posted by OpenADMET. Three model training methods were used in multitask mode to train models to predict pIC50 values. Multiple features were tested for their impact on the model performance.

The direct inhibition data for each of the four CYP's were split into five folds using Butina. The same folds were used for the ensuing model training.

1) A first model was trained using Uni-Mol v2 (168m weights), both in single task and multi-tasks, depending on the endpoint.

2) A D-MPNN-based model was trained using Chemprop in MTL with data augmentation. The four main tasks above were supplemented with auxiliary tasks:

- TDC Veith (PubChem AID 1851).
- Single dose screening data as log2FC as provided.
- Curated literature as available from ChEMBL 37.

Features, such LogD, pKSOL, HLM, MLM, and the physics-based Jazzy descriptors, and others improved the model performance. The use of fingerprint-derived spectrophores and RDKit 2D descriptors did not improve the model performance.

3) A third model was trained using LightGBM with ECFP4. To augment the data, the binding mode was determined for the training and blind set molecules using Protenix. The molecule poses were curated using Posebsters and further processed to extract PLIFs and signature of coordination to the heme's iron. These features were included with the model training.

Predictions for the blind set using the above three methods were then reduced to a single set using the Caruana bagged ensemble method. Finally, a Gaussian-based affine model was fitted and applied to each of the four endpoints, as inspired by a blog post by briford. CYP2D6 predictions showed the most significant shift.

Through the optimization of the workflow, it quickly became apparent that each of the four endpoints needed a separate set of models and features. So each endpoint had its own tailored process.